COMPUTER NETWORK PORTS

Definition, port ranges, service assignments and practical concepts

1. Definition of a Port

A **port** is a communication endpoint used by applications and services to send and receive data over a network. An **IP address** identifies the computer, while a **port number** identifies the application endpoint on that computer.

Example: 192.168.1.10:80 means the computer at IP address 192.168.1.10 is being contacted through port 80.

2. Types of Ports

The port-number range is divided into three main categories:

Port type	Port range	Definition	Examples
Well-known	0–1023	Used for common, widely standardized network services.	FTP 21, SSH 22, DNS 53, HTTP 80, HTTPS 443
Registered	1024–49151	Officially assigned to particular applications and services.	MS SQL 1433, MySQL 3306, RDP 3389, PostgreSQL 5432
Dynamic / Private	49152–65535	Generally used for temporary connections and private application communication.	50000, 52000, 60000

3. Why Are Ports Registered?

Companies and developers create network applications and services that need to communicate with other computers. Standard port assignments give clients and administrators a predictable place to connect.

• **MySQL** uses port 3306 by default.

• **PostgreSQL** uses port 5432 by default.

• **Microsoft SQL Server** commonly uses port 1433.

Many applications have a default port number hardcoded or configured in their software. This lets a client know which port to try without having to discover it every time.

Example: A MySQL client commonly connects to a MySQL server on port 3306. IANA maintains the official registry to provide standard assignments and help reduce conflicting assignments.

4. Other Important Concepts

4.1 Can we run any service on any port?

Yes. We can run almost any service on an available port, regardless of the service name normally associated with that port. For example, port 80 is commonly used for HTTP, but an SSH server or a custom application can be configured to use port 80 instead.

4.2 What if a port is already occupied?

Another application normally cannot use the same IP address, transport protocol and port while the existing application is listening there. You can stop or reconfigure the existing service, or choose a different port.

Example command to stop Apache on a typical Kali/Debian system:

```
sudo systemctl stop apache2
```

After Apache stops, another application may be able to use port 80.

4.3 Can two services run on the same port?

Normally, two services cannot listen on the exact same IP address, transport protocol and port at the same time. Advanced arrangements can let multiple backend services share one external port, for example:

• **Reverse proxy:** Nginx listens on port 80 and forwards requests to different backend applications.

• **Load balancing:** A front-end service distributes incoming connections or requests across multiple servers.

• **Socket sharing:** Special operating-system options can allow processes to share a listening socket.

• **Different IP addresses:** Two applications can use the same port number if they bind to different local IP addresses.

4.4 Can TCP and UDP use the same port number simultaneously?

Yes. TCP and UDP have separate port namespaces. Therefore, a computer can have a TCP service and a UDP service using the same numeric port at the same time, provided both endpoints are available and permitted.

Computer: 192.168.1.10	Application
TCP port 53	Application A
UDP port 53	Application B

This is possible across all three port ranges. For example, TCP 53 and UDP 53 can operate simultaneously. The operating system distinguishes them by transport protocol.

Important distinction: TCP port 53 and UDP port 53 are separate endpoints. Two ordinary TCP services generally cannot both bind the same IP and TCP port without special sharing or proxy arrangements.

4.5 Do we need to register a port to use it?

No. You do not need IANA registration to use an available port for a private or custom application. Official registration is useful when a protocol or service needs a standardized port assignment for broader use.

4.6 What determines which service is actually running?

The application listening on the port determines the service. The port number alone does not force a particular protocol or application to run there. A web server can use port 9000, for example, and a different application could use port 80.

5. Useful Port-Checking Commands

Command	Purpose
sudo ss -tulnp	Show listening TCP/UDP ports and process information.
sudo lsof -i :80	Find processes using port 80.
nc -zv 127.0.0.1 9000	Check connectivity to port 9000 on the local computer.

6. Final Points to Remember

• An IP address identifies the computer; a port identifies the application endpoint.

• Well-known ports are assigned to common standard services.

• Registered ports are assigned to particular applications and services.

• Dynamic/private ports are commonly used for temporary and private communication.

• Almost any service can use any available port if it is configured to do so.

• If a port is occupied, you normally need to stop or reconfigure the existing service.

• TCP and UDP can use the same port number simultaneously across all three port ranges.

• Advanced configurations such as reverse proxies can let multiple backend services share one external port.

Main concept: Port numbers provide standardized communication endpoints, but they do not force a particular application to run on a particular port.